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MODULAR PERSONAL CARE IMPLEMENT

Abstract

A modular personal care implement including a surface treatment head having a first end with an applicator and a second end including an extension; an end interconnector; and a sleeve positioned between the surface treatment head and the end interconnector, in which the extension extends through the sleeve and is received in the end interconnector such that the surface treatment head is releasably attached to the end interconnector. Also provided are a method of providing a modular personal care implement and a kit including a modular personal care implement.

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Background/Summary

FIELD OF THE INVENTION

[0001] The disclosure generally relates to a modular personal care implement, and more particularly to a modular personal care implement having a surface treatment head and at least one interconnector.

BACKGROUND OF THE INVENTION

[0002] Handles for personal care devices, such as toothbrushes and razors, are typically made from multiple components. These may include weighted components to give the razor a more substantial feel in the user's hand and various gripping areas so the personal care device does not slip out of the user's hand in a wet environment. These component parts are in addition to the cartridge connection, which may be formed as an integral part of the handle or may be formed separately and attached to the remainder of the handle.

[0003] Personal care device manufacturers have developed numerous assembly methods utilizing various handle components. Numerous different methods and styles of handles may be used in order to stratify the appeal of a growing population of diverse consumers. Numerous different assembly methods and components for different handle designs can be costly and time consuming to develop. In addition, consumers typically change their handle as new designs are released to the market. Accordingly, it would be desirable to provide a personal care implement that can be customized to accommodate the preferences of different users.

SUMMARY OF THE INVENTION

[0004] In accordance with an aspect of the present disclosure, a modular personal care implement is provided that includes a surface treatment head having a first end with an applicator and a second end comprising an extension; an end interconnector; and a sleeve positioned between the surface treatment head and the end interconnector, in which the extension extends through the sleeve and is received in the end interconnector such that the surface treatment head is releasably attached to the end interconnector.

[0005] In accordance with another aspect of the present disclosure, a method of providing a modular personal care implement is provided, in which the method includes: providing one or more surface treatment heads each having a first end with an applicator and a second end including an extension; providing a sleeve; providing an end interconnector; releasably attaching one surface treatment head of the one or more surface treatment heads to the end interconnector such that (i) the sleeve is positioned between the one surface treatment head and the end interconnector, and (ii) the surface treatment head is releasably attached to the end interconnector.

[0006] In accordance with a further aspect of the present disclosure, a kit is provided that includes a first surface treatment head comprising a shaving razor cartridge, the first surface treatment head having a first end with an applicator and a second end including an extension, in which the applicator includes at least one blade; a sleeve; an end interconnector; a case including a top portion and a bottom portion; and an insert including one or more cavities configured to receive the first surface treatment head, sleeve, and end interconnector. The bottom portion of the case defines a compartment configured to removably receive the insert within the bottom portion. The sleeve is separate from the first surface treatment head and the end interconnector, and a portion of an outer surface of the extension is configured to engage a portion of an inner surface of the end interconnector to form one of a mechanical connection or a magnetic connection that releasably attaches the surface treatment head to the end interconnector.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] While the specification concludes with claims particularly pointing out and distinctly claiming the subject matter which is regarded as forming the present invention, it is believed that

the invention will be better understood from the following description which is taken in conjunction with the accompanying drawings in which like designations are used to designate substantially identical elements, and in which:

[0008] FIG. **1** is a perspective view of a modular personal care implement, in accordance with the present disclosure.

[0009] FIG. **2** is an exploded view of the modular personal care implement of FIG. **1**.

[0010] FIG. **3A** is a cross-sectional view taken along line **3A-3A** in FIG. **1**.

[0011] FIG. **3B** is a detailed view of a portion of FIG. **3A**.

[0012] FIG. **4A** is an end view taken along view line **4A-4A** in FIG. **2**.

[0013] FIG. **4B** is an end view taken along view line **4B-4B** in FIG. **2**.

[0014] FIG. **5A** is an end view taken along view line **5A-5A** in FIG. **2**.

[0015] FIG. **5B** is a cross-sectional view taken along line **5B-5B** in FIG. **2**.

[0016] FIG. **6** is an end view taken along view line **6-6** in FIG. **2**.

[0017] FIG. **7** is a perspective view of a kit comprising a modular personal care implement, in accordance with the present disclosure.

[0018] FIG. **8** is an exploded view of a modular personal care implement, in accordance with the present disclosure.

[0019] FIG. **9A** is a cross-sectional view similar to FIG. **3A** of the modular personal care implement of FIG. **8**.

[0020] FIG. **9B** is a detailed view of a portion of FIG. **9A**.

[0021] FIG. **10** is an exploded view of a modular personal care implement, in accordance with the present disclosure.

[0022] FIG. **11A** is a cross-sectional view similar to FIG. **3A** of the modular personal care implement of FIG. **10**.

[0023] FIG. **11B** is a detailed view of a portion of FIG. **11A**.

[0024] FIG. **12** is a flow diagram of a method of providing a modular personal care implement, in accordance with the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0025] With reference to FIGS. **1**, **2**, and **3A**, a modular personal care implement **10** in accordance with the present disclosure is illustrated. The modular personal care implement **10** comprises a handle section **12** and a surface treatment head **14**. The modular personal care implement **10** may be a handheld personal grooming implement, such as a shaving razor, a toothbrush, and the like. The surface treatment head **14** has a first end **16** with an applicator **18**, and a second end **20** comprising an extension **40**. The surface treatment head **14** may comprise a first portion **14A** that includes the applicator **18**, and a second portion **14B** that includes the extension **40**. The surface treatment head **14** may further comprise a release mechanism positioned between the first and second ends **16** and **20**, i.e., between the first and second portions **14A** and **14B**. The release mechanism may comprise an actuator button **50** configured to eject the first portion **14A** comprising the applicator **18** from the second portion **14B** comprising the extension **40**.

[0026] With reference to FIG. **2**, the first portion **14A** of the surface treatment head **14** may comprise a shaving razor cartridge **24**, and the applicator **18** may comprise at least one blade **26** (see FIG. **3A**). The shaving razor cartridge **24** may comprise a base **52** with two arms **54** that pivotally support a housing (not separately labeled), in which the housing includes the at least one blade **26**. The second portion **14B** of the surface treatment head **14** may comprise a cartridge support structure **56** with an extension **58** and a spring-biased plunger **60**. The base **52** of the shaving razor cartridge **24** includes a cavity **62** that is configured to receive the extension **58** and the spring-biased plunger **60**. The actuator button **50** may be moved toward the shaving razor cartridge **24**, and the spring-biased plunger **60** causes the shaving razor cartridge **24** to be ejected from the cartridge support structure **56**.

[0027] Although the surface treatment head **14** is depicted as a shaving razor cartridge **24** in FIGS.

1, 2, and 3A, it is understood that it may alternatively be a brush head having bristles, e.g., a toothbrush or other personal grooming or hygiene product. For example, with reference to FIG. 7, the modular personal care implement **10** may comprise a surface treatment head **14'** that may comprise a first end **16'** with an applicator **18'** and a second end **20'** with an extension **40'** that may be substantially similar to the extension **40** of the surface treatment head **14** comprising the shaving razor cartridge **24**. The second surface treatment head **14'** may comprise a brush head **28**, e.g., a toothbrush, and the applicator **18'** comprises a plurality of bristles **84**.

[0028] As shown in FIGS. 1, 2, and 3A, the handle section **12** of the modular personal care implement **10** comprises a sleeve **30** and an end interconnector **32**. The sleeve **30** has a first end **34** and a second end **36**. The sleeve **30** comprises a hollow structure with an opening **46** extending from the first end **34** to the second end **36** (see also FIGS. 4A and 4B). The end interconnector **32** has a first end **42** and a second end **44**, in which the second end **44** of the end interconnector **32** may define a terminus or terminal end of the handle section **12**.

[0029] To assemble the modular personal care implement **10**, the sleeve **30** is placed over the extension **40**, such that the extension **40** is received in and extends through the opening **46** of the sleeve **30**. An inner surface **31** of the sleeve **30** comprises a shape, i.e., a substantially semicircular or hemispherical cross-sectional shape, that corresponds to an adjacent portion of an outer surface **41** of the extension **40**, such that the extension **40** fits snugly within the sleeve **30**, as shown in FIGS. 3A and 3B. As described herein, the extension **40** is then inserted into and received in the end interconnector **32**, such that the surface treatment head **14** is releasably attached to the end interconnector **32**. Upon assembly, the sleeve **30** is positioned between the surface treatment head **14** and the end interconnector **32**. As used herein, the term “releasably attach” and similar terms are intended to refer to a connection or coupling that is configured to be formed repeatedly by a consumer attaching and detaching components under normal use conditions without causing damage to the components.

[0030] As shown in FIGS. 3A, 3B, and 5A, the end interconnector **32** comprises a bore **48** extending longitudinally inward from the first end **42** toward the second end **44**. The bore **48** comprises an inner surface **49** with a shape, i.e., a substantially semicircular or hemispherical cross-sectional shape, that corresponds to an adjacent portion of the outer surface **41** of the extension **40**, such that the extension **40** fits snugly within the bore **48**, as shown in FIGS. 3A and 3B. As described herein, a portion of the outer surface **41** of the extension **40** engages a portion of an inner surface of the end interconnector **32**, i.e., a portion of the inner surface **49** of the bore **48**, to form one of a mechanical connection or a magnetic connection that releasably attaches the surface treatment head **14** to the end interconnector **32**. Upon assembly, the sleeve **30** is held in position between the surface treatment head **14** and the end interconnector **32** by engagement between the portion of the outer surface **41** of the extension **40** and the portion of the inner surface of the end interconnector **32**, i.e., the portion of the inner surface **49** of the bore **48**.

[0031] In the example shown in FIGS. 2, 3A, and 3B, a connection between the extension **40** and the end interconnector **32** may comprise a mechanical connection in the form of a ball-snap mechanism. The portion of the outer surface **41** of the extension **40** that engages the portion of the inner surface of the end interconnector **32** may comprise a spring-loaded ball mechanism **64**. As shown in FIGS. 2, 3B, and 6, the outer surface **41** of the extension **40** may comprise a planar surface **72** and a curved surface **74** opposite the planar surface **72**, in which the planar and curved surfaces **72** and **74** define a substantially semicircular or hemispherical cross-sectional shape that corresponds to the shape of the inner surface **31** of the sleeve **30** and the inner surface **49** of the bore **48**. The planar surface **72** comprises a cavity **43** extending inward toward the curved surface **74**, and the spring-loaded ball mechanism **64** may be inserted into and fixed to the cavity **43**, e.g., by a press-fitting process and/or gluing. The spring-loaded ball mechanism **64** may comprise a spring **68** and a spring-loaded ball **70**, in which the spring **68** may apply a force onto the ball **70** that biases the ball **70** outward, such that the ball **70** extends slightly above the outer surface **41** of

the extension **40** as shown in FIG. 6. The ball **70** and/or the spring **68** may comprise one or more materials, including a plastic and/or a metal, such as stainless steel.

[0032] As shown in FIGS. 3B and 6, the portion of the inner surface of the end interconnector **32**, i.e., a portion of the inner surface **49** of the bore **48**, comprises a recess **66** that is configured to receive the ball **70**. When the modular personal care implement **10** is assembled, the extension **40** may be fed linearly through the sleeve **30** and into the bore **48** formed in the end interconnector **32**. The spring **68** allows the ball **70** to drop down into the cavity **43** so that the ball **70** is flush with the outer surface **41** of the extension **40** as the extension **40** passes through the sleeve **30** and into the bore **48**. When the extension **40** is received in the end interconnector **32**, the spring **68** biases the ball **70** outward so that the ball **70** engages the recess **66**, as best seen in FIGS. 3A and 3B. Engagement between the ball **70** and the recess **66** provides a snap-fit locking mechanism to ensure sufficiently strong connection and stability between the surface treatment head **14** and the end interconnector **32**, e.g. to enable a user to perform a shaving or brushing action. Once the extension **40** is snap-fitted into the end interconnector **32**, the surface treatment head **14** is fixed in an axial direction so that the surface treatment head **14** does not loosen, twist, or rotate during use. The snap-fit may provide a precise fixation of the surface treatment head **14**, and a distinct haptic feedback may be given to the user that the surface treatment head **14** is snapped-on securely, i.e., the user may recognize once the ball **70** engages into the recess **66**. The surface treatment head **14** may be fixed to the end interconnector **32** until a specific/predetermined pull-off force is applied. The surface treatment head **14** may be easily removed by applying a linear motion to pull the surface treatment head **14** apart from the end interconnector **32**. When the predetermined pull-off force is exceeded, the ball **70** is forced back into the cavity **43**, and the extension **40** may be removed from the end interconnector **32** and the sleeve **30**.

[0033] With reference to FIGS. 2, 3A, 3B, and 6, the extension **40** may comprise a tab **40A** extending outward therefrom. The tab **40A** may mirror a shape of the extension **40** and may comprise a planar surface **72A** and a curved surface **74A** opposite the planar surface **72A**. A width $W_{sub.4}$ of the tab **40A** may be less than a width $W_{sub.1}$ of the extension **40**, in which the planar surface **72** of the extension **40** and the planar surface **72A** of the tab **40A** extend within a same plane (not shown) and the curved surface **74A** of the tab **40A** is recessed with respect to the curved surface **74** of the extension **40**. The end interconnector **32** may comprise a further bore **48A** extending longitudinally inward from the bore **48** toward the second end **44**. When the extension **40** is received in the end interconnector **32**, the tab **40A** is received in the further bore **48A**, which may serve as, for example, a location feature to ensure that the surface treatment head **14** is properly aligned with the end interconnector **32**.

[0034] As shown in FIGS. 4A, 4B, 5A, and 5B, an outer surface **30-1** of the sleeve **30** may comprise a plurality of gripping protrusions **76**, and an outer surface **32-1** of the end interconnector **32** may be smooth. The gripping protrusions **76** of the sleeve **30** may assist with gripping the modular personal care implement **10** in a wet environment.

[0035] The sleeve **30** may comprise a first cross-sectional shape, and at least a portion of the end interconnector **32** may comprise a second cross-sectional shape that is different from the first cross-sectional shape. For example, as shown in FIGS. 1, 3A, 4A, and 4B, the outer surface **30-1** of the sleeve **30** (excluding the gripping protrusions **76**) defines a substantially semicircular or hemispherical cross-sectional shape with a flat portion and a curved portion, in which the shape of the sleeve **30** may be uniform along an entirety of a length (not labeled) of the sleeve **30**. As shown in FIGS. 1, 3A, 5A, and 5B, the outer surface **32-1** of the end interconnector **32** near the first end **42** defines a substantially semicircular or hemispherical cross-sectional shape that corresponds to the cross-sectional shape of the sleeve **30** shown in FIGS. 1, 4A, and 4B. The semicircular/hemispherical cross-sectional shape of the sleeve **30** and the first end **42** of the end interconnector **32** may provide an ergonomic shape that may assist with gripping the modular personal care implement **10**. The end interconnector **32** may include a transition section **78** located

between the first and second ends **42** and **44**, in which the cross-sectional shape of the end interconnector **32** transitions from the substantially semicircular/hemispherical cross-sectional shape near the first end **42** to a substantially circular cross-sectional shape near the second end **44**.

[0036] The surface treatment head **14**, the sleeve **30**, and the end interconnector **32** may comprise one or more materials and may have one or more different properties such as hardness, coefficient of friction, surface texture, mass, color, length, and/or width. The sleeve **30** may comprise one or more first materials, and the end interconnector **32** may comprise one or more second materials that are the same as or different from the first material(s). For example, the sleeve **30** and the end interconnector **32** may comprise one or more elastomers, silicones, engineering plastics, or metals.

[0037] In some examples, the end interconnector **32** may comprise a relatively heavy material such as a metal or metal alloy and/or the end interconnector **32** may optionally comprise an insert (not shown) to increase an overall mass of the modular personal care implement **10**. In some examples, the overall mass of the modular personal care implement **10** may be about 15 g. The surface treatment head **14**, including the extension **40**, may comprise the same or different material(s).

[0038] The sleeve **30** may comprise a material with a first hardness, and the end interconnector **32** may comprise a material with a second hardness that is the same as or different from the first hardness. In some examples, the sleeve **30** and/or end interconnector **32** may comprise a softer material with a shore A hardness of about 20 to 40. In other examples, the sleeve **30** and/or end interconnector **32** may comprise a harder material with a shore A hardness greater than 50, e.g., from about shore A 60 to about shore D 100. Materials comprising metal may have a shore D hardness of greater than 100. The material(s) comprising the surface treatment head **14** may have the same hardness or a different hardness. Softer materials may provide a more comfortable and improved gripping surface, while harder materials may provide greater durability and resistance to damage.

[0039] The sleeve **30** may comprise a material with a first coefficient of friction (CoF), and the end interconnector **32** may comprise a material with a second CoF that is the same as or different from the first CoF. The material(s) comprising the surface treatment head **14** may have the same CoF or a different CoF. Materials with a higher CoF may provide a better gripping surface, e.g., within the handle section **12**, while materials with a lower CoF provide higher glide or slipperiness.

[0040] The sleeve **30** may comprise a first surface texture, and the end interconnector **32** may comprise a second surface texture that is the same as or different from the first surface texture. The surface treatment head **14** may have the same surface texture or a different surface texture. The surface texture may be determined, at least in part, by the material(s) and/or the presence or absence of gripping protrusions **76** and/or other structures that form one or more protrusions or peaks and/or recesses or valleys on a surface of the component.

[0041] The sleeve **30** may comprise one or more first colors, and the end interconnector **32** may comprise one or more second colors that are the same as or different from the first color(s). The surface treatment head **14** may have the same color(s) or one or more different colors. For example, the surface treatment head **14**, the sleeve **30**, and/or the end interconnector **32** may be selected to include a color combination corresponding to a country flag, a favorite sports team, etc.

[0042] The second portion **14B** of the surface treatment head **14** (specifically the extension **40**), the sleeve **30**, and the end interconnector **32** may each comprise a respective length $L_{sub.1}$, $L_{sub.2}$, and $L_{sub.3}$ and a respective width $W_{sub.1}$, $W_{sub.2}$, and $W_{sub.3}$. The length(s) $L_{sub.1}$, $L_{sub.2}$, and $L_{sub.3}$ and/or widths $W_{sub.1}$, $W_{sub.2}$, and $W_{sub.3}$ of the components may be varied as desired to obtain a desired configuration of the handle section **12**. In some examples, the overall length of the handle section **12** may be 120 mm to 150 mm. In other examples, the overall length of the handle section **12** may be 600 mm or greater, e.g., for hard-to-reach areas such as the user's back.

[0043] As shown in FIG. 7, the modular personal care implement **10** in accordance with the present disclosure may be part of a kit **80** comprising multiple components, including one or more surface

treatment heads **14** and **14'**, a sleeve **30**, and an end interconnector **32**. In the example shown in FIG. 7, the kit **80** includes a first surface treatment head **14** comprising a shaving razor cartridge **24**, a second surface treatment head **14'** comprising a brush head **28**, and an end interconnector **32**. It may be understood that the kit **80** may include any number or configuration of the components described herein, e.g., one or more additional sleeves **30** and/or end interconnectors **32**. In some examples, the kit **80** may include two or more components having a different length and/or width (see FIG. 3A) so that an overall length and/or width (not labeled) of the handle section **12** may be customized as desired. For example, the kit **80** may include two surface treatment heads having extensions with different lengths and/or widths (not shown); two sleeves with different lengths and/or widths (not shown); and/or two end interconnectors having extensions with different lengths and/or widths (not shown).

[0044] The kit **80** may include a case **82** comprising a top portion **86** and a bottom portion **94** that is configured to removably receive an insert **88** within the bottom portion **94**. The insert **88** may comprise one or more cavities **90** configured, i.e., sized and shaped, to receive the components of the kit **80**, e.g., the surface treatment head(s) **14** and/or **14'**, the sleeve **30**, and the end interconnector **32**. The bottom portion **94** may be configured to be received in the top portion **86** to enclose the kit **80** and the components thereof. The top and bottom portions **86** and **94** may be coupled together by a friction fit, a hinge (not shown), and/or by other means, such as one or more adhesive stickers or labels. The top and bottom portions **86** and **94** of the case **82** may comprise a durable material such as an engineering plastic (e.g., a polycarbonate), a metal, or a metal alloy (e.g., steel, aluminum, tin, or combinations thereof). The case **82** may provide a sturdy, multi-use container that protects the components of the kit **80**. The case **82** may also serve as a durable and convenient storage and travel kit and may be used to hold, organize, and store the components of the modular personal care implement **10** for day-to-day use. The case **82** may comprise a hang tag **92** that may be coupled to the case **82** via adhesive or other suitable attachment means or may be formed as part of the case **82**. The hang tag **92** may be removable. The hang tag **92** may be configured to allow the kit **80** to be displayed and/or stored and may comprise an aperture (not labeled) that allows the kit **80** to be associated with a display and/or storage system (not shown) by, for example, hanging the kit **80** from a hook or peg.

[0045] The modular personal care implement **10** in accordance with the present disclosure provides an ergonomic handle section **12** with separable components that allow a user to customize the configuration as desired and to easily replace the components or a portion thereof. With reference to FIGS. 1, 2, and 3A, the surface treatment head **14** comprising the shaving razor cartridge **24** may be replaced in its entirety by detaching the extension **40** from the end interconnector **32** and releasably attaching a new surface treatment head **14**. Alternatively, only the shaving razor cartridge **24** may be replaced by actuating the actuator button **50** to eject the shaving razor cartridge **24** from the cartridge support structure **56** and attaching a new shaving razor cartridge **24** to the cartridge support structure **56**. The surface treatment head **14'** comprising a brush head **28** (see FIG. 7) may be replaced by detaching the extension **40'** from the end interconnector **32** and releasably attaching a new surface treatment head **14'**. The sleeve **30** and/or the end interconnector **32** may also be detached and replaced. The user may select a combination of the surface treatment head **14**, **14'**, the sleeve **30**, and/or the end interconnector **32** having desired lengths and/or widths to customize the overall length and/or width of the handle section **12** as desired.

[0046] FIGS. 8, 9A, and 9B illustrate another example of a modular personal care implement **100** in accordance with the present disclosure. The modular personal care implement **100** comprises a handle section **112** and a surface treatment head **114**. The surface treatment head **114** has a first end **116** with an applicator **118**, and a second end **120** with an extension **140**. Apart from the extension **140**, the surface treatment head **114** may be substantially similar to the surface treatment head **14** described with respect to FIGS. 1, 2, 3A, and 3B and may comprise a first portion **114A** that includes the applicator **118** and a second portion **114B** that includes the extension **140**. As

described with respect to the modular personal care implement **10**, the surface treatment head **114** may further comprise a release mechanism, e.g., an actuator button **150**, that is configured to eject the first portion **114A** comprising the applicator **118** from the second portion **114B** comprising the extension **140**. In the example shown in FIG. **8**, the surface treatment head **114** comprises a shaving razor cartridge **124**, and the applicator **118** comprises at least one blade **126**. It may be understood that the surface treatment head **114** may comprise a brush head, as described with respect to FIG. **7**. [0047] The handle section **112** of the modular personal care implement **100** comprises a sleeve **130** and an end interconnector **132**. The sleeve **130** has a first end **134** and a second end **136** and comprises a hollow structure with an opening **146** extending from the first end **134** to the second end **136**. The sleeve **130** may be substantially similar to the sleeve **30** described herein except that the opening **146** is circular instead of semicircular or hemispherical. Apart from a connection between the extension **140** and the end interconnector **132**, the end interconnector **132** may be substantially similar to the end interconnector **32** described herein. The end interconnector **132** has a first end **142** and a second end **144**, in which the second end **144** of the end interconnector **132** may define a terminus or terminal end of the handle section **112**.

[0048] As shown in FIGS. **9A** and **9B**, the end interconnector **132** comprises a bore **148** extending longitudinally inward from the first end **142** toward the second end **144**. A portion of an outer surface **141** of the extension **140** engages a portion of an inner surface of the end interconnector **132**, i.e., a portion of an inner surface **149** of the bore **148**, to form a mechanical connection that releasably attaches the surface treatment head **114** to the end interconnector **132**. The extension **140** may comprise a circular cross-section, such that the extension **140** fits within the sleeve **130**. Similar to the modular personal care implement **10**, upon assembly of the modular personal care implement **100**, the sleeve **130** is held in position between the surface treatment head **114** and the end interconnector **132** by engagement between the portion of the outer surface **141** of the extension **140** and the portion of the inner surface of the end interconnector **132**, i.e., the portion of the inner surface **149** of the bore **148**.

[0049] In the example shown in FIGS. **8**, **9A**, and **9B**, the connection between the extension **140** and the end interconnector **132** may comprise a mechanical connection in the form of a threaded connection. The portion of the outer surface **141** of the extension **140** that engages the portion of the inner surface of the end interconnector **132** may comprise a tab **140A** extending outward therefrom, in which the tab **140A** comprises external threads **164**. The tab **140A** may comprise a circular cross-section. The portion of the inner surface **149** of the bore **148** also comprises a circular cross-section and includes corresponding internal threads **168** that receive the external threads **164** to securely and releasably attach the surface treatment head **114** to the end interconnector **132**. The modular personal care implement **100** may be disassembled by unscrewing the surface treatment head **114** from the end interconnector **132**. In the example shown in FIGS. **9A** and **9B**, only the tab **140A** is received in the bore **148**. In other examples (not shown), the bore **148** may also receive a portion of the extension **140**, similar to the example shown in FIGS. **3A** and **3B**.

[0050] The modular personal care implement **100** may be part of a kit (not shown) comprising multiple components, including one or more surface treatment heads **114**, one or more sleeves **130**, and at least one end interconnector **132**. The kit comprising the modular personal care implement **100** may be substantially similar to the kit **80** described with respect to FIG. **7**. As described with respect to the modular personal care implement **10**, the modular personal care implement **100** provides an ergonomic handle section **112** that allows a user to customize the configuration as desired, e.g., to obtain a desired length and/or width (not labeled) of the handle section **112** and/or to adjust the hardness, coefficient of friction, surface texture, mass, and/or color to obtain a desired look and feel for the handle section **112**. The surface treatment head **114** may also be easily replaced, as described herein.

[0051] FIGS. **10**, **11A**, and **11B** illustrate another example of a modular personal care implement **200** in accordance with the present disclosure. The modular personal care implement **200**

comprises a handle section **212** and a surface treatment head **214**. The surface treatment head **214** has a first end **216** with an applicator **218**, and a second end **220** with an extension **240**. Apart from the extension **240**, the surface treatment head **214** may be substantially similar to the surface treatment head **14** described herein and may comprise a first portion **214A** that includes the applicator **218** and a second portion **214B** that includes the extension **240**. As described with respect to the modular personal care implement **10**, the surface treatment head **214** may further comprise a release mechanism, e.g., an actuator button **250**, that is configured to eject the first portion **214A** comprising the applicator **218** from the second portion **214B** comprising the extension **240**. In the example shown in FIG. **10**, the surface treatment head **214** comprises a shaving razor cartridge **224**, and the applicator **218** comprises at least one blade **226**. It may be understood that the surface treatment head **214** may comprise a brush head, as described with respect to FIG. **7**.

[0052] The handle section **212** of the modular personal care implement **200** comprises a sleeve **230** and an end interconnector **232**. The sleeve **230** has a first end **234** and a second end **236** and comprises a hollow structure with an opening (not labeled) extending from the first end **234** to the second end **236**. The sleeve **230** may be identical or substantially similar to the sleeve **30** described herein, and an inner surface (not labeled) of the sleeve **230** comprises a shape, i.e., a substantially semicircular or hemispherical cross-sectional shape, that corresponds to an adjacent portion of an outer surface **241** of the extension **240**, such that the extension **240** fits snugly within the sleeve **230**. Apart from a connection between the extension **240** and the end interconnector **232**, the end interconnector **232** may be substantially similar to the end interconnector **32** described herein. The end interconnector **232** has a first end **242** and a second end **244**, in which the second end **244** of the end interconnector **232** may define a terminus or terminal end of the handle section **212**.

[0053] As shown in FIGS. **11A** and **11B**, the end interconnector **232** comprises a bore **248** extending longitudinally inward from the first end **242** toward the second end **244**. The bore **248** comprises an inner surface **249** with a shape, i.e., a substantially semicircular or hemispherical cross-sectional shape, that corresponds to an adjacent portion of the outer surface **241** of the extension **240**, such that the extension **240** fits snugly within the bore **248**. A portion of the outer surface **241** of the extension **240** engages a portion of an inner surface of the end interconnector **232**, i.e., a portion of an inner surface **249** of the bore **248**, to form a connection that releasably attaches the surface treatment head **214** to the end interconnector **232**. Similar to the modular personal care implement **10**, upon assembly of the modular personal care implement **200**, the sleeve **230** is held in position between the surface treatment head **214** and the end interconnector **232** by engagement between the portion of the outer surface **241** of the extension **240** and the portion of the inner surface of the end interconnector **232**, i.e., the portion of the inner surface **249** of the bore **248**.

[0054] In the example shown in FIGS. **10**, **11A**, and **11B**, the connection between the extension **240** and the end interconnector **232** may comprise a magnetic connection. The portion of the outer surface **241** of the extension **240** comprises a magnetic material, e.g., a magnet **225**, and the portion of the inner surface of the end interconnector **232** comprises a magnetic material, e.g., a magnet **227**. The magnetic materials each produce a magnetic field, in which the magnetic material of the extension **240** is attracted to the magnetic material of the end interconnector **232** to create the magnetic connection and securely and releasably attach the surface treatment head **214** to the end interconnector **232**. The substantially semicircular or hemispherical cross-sectional shape of the extension **240** and the bore **248** may help to prevent rotation of the surface treatment head **214** and align the surface treatment head **214** with respect to the end interconnector **132**. The modular personal care implement **200** may be disassembled by pulling the surface treatment head **214** away from the end interconnector **232** with a force sufficient to break the magnetic connection.

[0055] The modular personal care implement **200** may be part of a kit (not shown) comprising multiple components, including one or more surface treatment heads **214**, one or more sleeves **230**,

and at least one end interconnector **232**. The kit comprising the modular personal care implement **200** may be substantially similar to the kit **80** described with respect to FIG. 7. As described with respect to the modular personal care implement **10**, the modular personal care implement **200** provides an ergonomic handle section **212** that allows a user to customize the configuration as desired, e.g., to obtain a desired length and/or width (not labeled) of the handle section **212** and/or to adjust the hardness, coefficient of friction, surface texture, mass, and/or color to obtain a desired look and feel for the handle section **212**. The surface treatment head **214** may also be easily replaced, as described herein.

[0056] FIG. **12** is a flow diagram of a method **300** of providing a modular personal care implement, in accordance with the present disclosure. As shown in FIG. **12**, the method **300** includes providing one or more surface treatment heads each having a first end with an applicator and a second end comprising an extension at **310**; providing a sleeve at **320**; providing an end interconnector at **330**; and at **340**, releasably attaching one surface treatment head of the one or more surface treatment heads to the end interconnector such that (i) the sleeve is positioned between the one surface treatment head and the end interconnector, and (ii) the surface treatment head is releasably attached to the end interconnector, after which the method **300** may conclude.

[0057] The one surface treatment head may comprise a first applicator, and the method may further comprise: removing the one surface treatment head from the end interconnector; selecting a second surface treatment head comprising a second applicator; and releasably attaching the second surface treatment head to the end interconnector such that (i) the sleeve is positioned between the second surface treatment head and the end interconnector, and (ii) the second surface treatment head is releasably attached to the end interconnector.

[0058] The modular personal care implement formed by the method **300** may include any of the modular personal care implements **10**, **100**, and **200** described herein. Releasably attaching the one surface treatment head of the one or more surface treatment heads to the end interconnector may comprise engaging a portion of an outer surface of the extension with a portion of an inner surface of the end interconnector to form one of a mechanical connection or a magnetic connection, in which the sleeve is held in position between the one surface treatment head and the end interconnector by engagement between the portion of the outer surface of the extension and the portion of the inner surface of the end interconnector. In some examples, the mechanical connection may comprise a ball-snap connection, in which the portion of the outer surface of the extension may comprise a spring-loaded ball, the portion of the inner surface of the end interconnector comprises a recess, and the extension extends through the sleeve and is received in the end interconnector such that the spring-loaded ball engages the recess.

[0059] The one or more surface treatment heads may each further comprise a release mechanism positioned between the first and second end of the surface treatment head, and the method may further comprise actuating the release mechanism to eject a portion of the one surface treatment head comprising the applicator.

[0060] The dimensions and values disclosed herein are not to be understood as being strictly limited to the exact numerical values recited. Instead, unless otherwise specified, each such dimension is intended to mean both the recited value and a functionally equivalent range surrounding that value. For example, a dimension disclosed as “40 mm” is intended to mean “about 40 mm.”

[0061] Every document cited herein, including any cross referenced or related patent or application and any patent application or patent to which this application claims priority or benefit thereof, is hereby incorporated herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references, teaches, suggests, or discloses any such invention. Further, to the extent that any meaning or definition of a term in this document conflicts with any meaning or definition

of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

[0062] While particular embodiments of the present invention have been illustrated and described, it would be obvious to those skilled in the art that various other changes and modifications can be made without departing from the spirit and scope of the invention. It is therefore intended to cover in the appended claims all such changes and modifications that are within the scope of this invention.

Claims

1. A modular personal care implement comprising: a surface treatment head having a first end with an applicator and a second end comprising an extension; an end interconnector; and a sleeve positioned between the surface treatment head and the end interconnector, wherein the extension extends through the sleeve and is received in the end interconnector such that the surface treatment head is releasably attached to the end interconnector.
2. The modular personal care implement of claim 1, wherein a portion of an outer surface of the extension engages a portion of an inner surface of the end interconnector to form one of a mechanical connection or a magnetic connection that releasably attaches the surface treatment head to the end interconnector.
3. The modular personal care implement of claim 2, wherein the sleeve is held in position between the surface treatment head and the end interconnector by engagement between the portion of the outer surface of the extension and the portion of the inner surface of the end interconnector.
4. The modular personal care implement of claim 2, wherein the mechanical connection comprises a ball-snap connection, and wherein: the portion of the outer surface of the extension comprises a spring-loaded ball; the portion of the inner surface of the end interconnector comprises a recess; and the extension extends through the sleeve and is received in the end interconnector such that the spring-loaded ball engages the recess.
5. The modular personal care implement of claim 4, wherein the extension comprises a planar surface and a curved surface opposite the planar surface, the planar surface comprising a cavity that receives the spring-loaded ball and a spring.
6. The modular personal care implement of claim 2, wherein the mechanical connection comprises a threaded connection.
7. The modular personal care implement of claim 1, wherein the sleeve comprises a first cross-sectional shape and at least a portion of the end interconnector comprises a second cross-sectional shape that is different from the first cross-sectional shape.
8. The modular personal care implement of claim 1, wherein the sleeve comprises a first material and the end interconnector comprises a second material that is different from the first material.
9. The modular personal care implement of claim 1, wherein an outer surface of the sleeve comprises a plurality of gripping protrusions and an outer surface of the end interconnector is smooth.
10. The modular personal care implement of claim 1, wherein the surface treatment head comprises a shaving razor cartridge and the applicator comprises at least one blade.
11. The modular personal care implement of claim 1, wherein the surface treatment head comprises a brush head and the applicator comprises a plurality of bristles.
12. The modular personal care implement of claim 1, wherein the surface treatment head further comprises a release mechanism positioned between the first and second end of the surface treatment head, the release mechanism comprising an actuator button configured to eject a portion of the surface treatment head comprising the applicator.
13. A method of providing a modular personal care implement, the method comprising: providing one or more surface treatment heads each having a first end with an applicator and a second end

comprising an extension; providing a sleeve; providing an end interconnector; releasably attaching one surface treatment head of the one or more surface treatment heads to the end interconnector such that (i) the sleeve is positioned between the one surface treatment head and the end interconnector, and (ii) the surface treatment head is releasably attached to the end interconnector.

14. The method of claim 13, wherein the one surface treatment head comprises a first applicator, the method further comprising: removing the one surface treatment head from the end interconnector; selecting a second surface treatment head comprising a second applicator; and releasably attaching the second surface treatment head to the end interconnector such that (i) the sleeve is positioned between the second surface treatment head and the end interconnector, and (ii) the second surface treatment head is releasably attached to the end interconnector.

15. The method of claim 13, wherein releasably attaching the one surface treatment head of the one or more surface treatment heads to the end interconnector comprises engaging a portion of an outer surface of the extension with a portion of an inner surface of the end interconnector to form one of a mechanical connection or a magnetic connection, wherein the sleeve is held in position between the one surface treatment head and the end interconnector by engagement between the portion of the outer surface of the extension and the portion of the inner surface of the end interconnector.

16. The method of claim 15, wherein the mechanical connection comprises a ball-snap connection, and wherein: the portion of the outer surface of the extension comprises a spring-loaded ball; the portion of the inner surface of the end interconnector comprises a recess; and the extension extends through the sleeve and is received in the end interconnector such that the spring-loaded ball engages the recess.

17. The method of claim 13, wherein the one or more surface treatment heads each further comprise a release mechanism positioned between the first and second end of the surface treatment head, the method further comprising: actuating the release mechanism to eject a portion of the one surface treatment head comprising the applicator.

18. A kit comprising: a first surface treatment head comprising a shaving razor cartridge, the first surface treatment head having a first end with an applicator and a second end comprising an extension, wherein the applicator comprises at least one blade; a sleeve; an end interconnector; a case comprising a top portion and a bottom portion; and an insert comprising one or more cavities configured to receive the first surface treatment head, sleeve, and end interconnector, wherein the bottom portion of the case defines a compartment configured to removably receive the insert within the bottom portion; wherein the sleeve is separate from the first surface treatment head and the end interconnector; and wherein a portion of an outer surface of the extension is configured to engage a portion of an inner surface of the end interconnector to form one of a mechanical connection or a magnetic connection that releasably attaches the surface treatment head to the end interconnector.

19. The kit of claim 18, further comprising: a second surface treatment head comprising a brush head, the second surface treatment head having a first end with an applicator and a second end comprising an extension, wherein the applicator comprises a plurality of bristles; and wherein the insert further comprises at least one additional cavity configured to receive the second surface treatment head.

20. The kit of claim 18, wherein at least one of: the sleeve comprises a first cross-sectional shape and at least a portion of the end interconnector comprises a second cross-sectional shape that is different from the first cross-sectional shape; the sleeve comprises a first material and the end interconnector comprises a second material that is different from the first material; or an outer surface of the sleeve comprises a plurality of gripping protrusions and an outer surface of the end interconnector is smooth.
